Conceptual design of aircraft

Balance of forces (intro to flight mechanics)

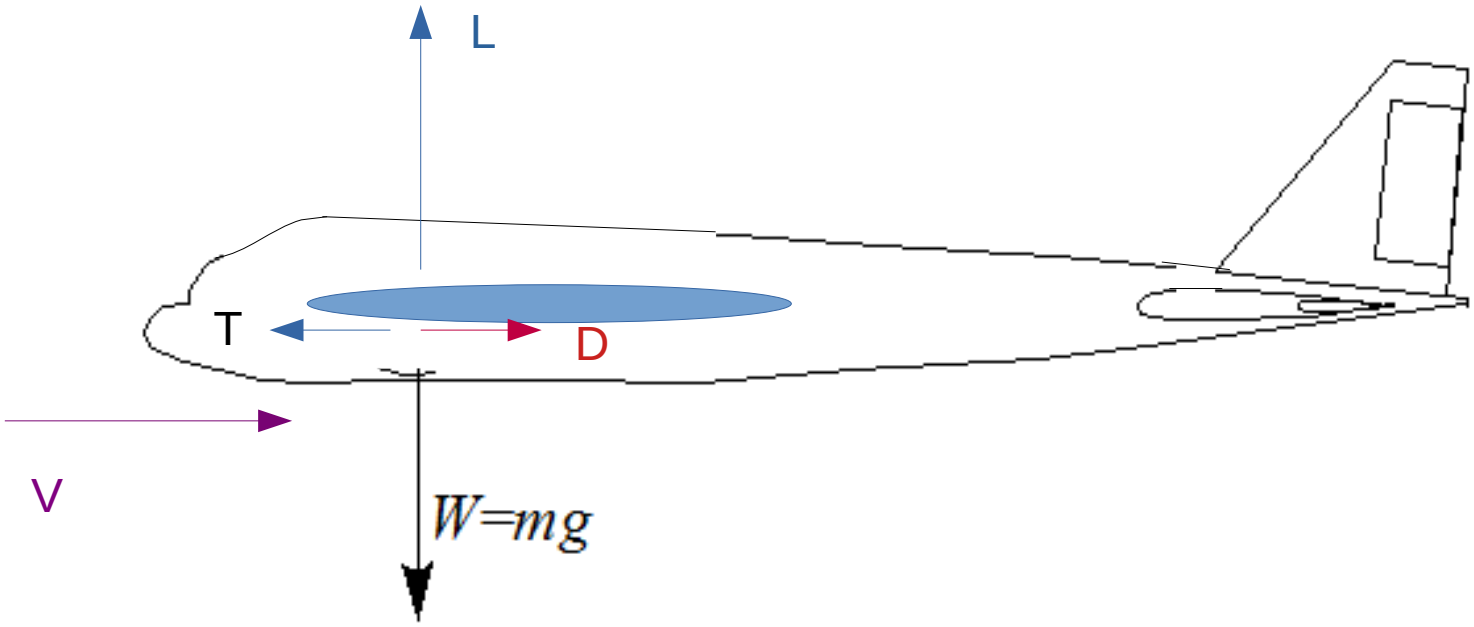
Objectives:

By the end of the session the student will be able to:

- Derive the equations governing the balance of forces for the following steady manoeuvres:
    - Cruise in steady level flight
    - Steady climb
    - Steady descent
- Analyze numerically simple situations of the above manoeuvres, e.g. finding aerodynamic coefficients and aircraft velocity.

Eldad Avital

Steady level flight



In steady level flight, V is constant and horizontal:

L = W, F = D, where: L is the lift, D is the drag, F is the thrust

Example 1:
Aircraft flies in standard sea level conditions with C_L=1. Its mass is 10000 kg and the wings surface area is 10 m^2. Calculate the flight velocity V.

Data:
m = 10000 (kg)
S = 10 (m^2)
C_L=1,
h=0 of standard conditions

Question:
V=?

Conceptual design of aircraft

Properties of the Standard Atmosphere

$p_o = 101\,325\ \mathrm{N/m^2}$ $T_o = 288.16\ \mathrm{K}$ $\mu_o = 1.783 \times 10^{-5}\ \mathrm{kg/ms}$
$\rho_o = 1.225\ \mathrm{kg/m^3}$ $a_o = 340.3\ \mathrm{m/s}$ $\nu_o = 1.455 \times 10^{-5}\ \mathrm{m^2/s}$

Height metres	Relative Temperature		Relative Pressure	Relative Density		Relative Viscosity	
h	θ	$\sqrt{\theta}$	ϖ	σ	$\sqrt{\sigma}$	μ/μ_o	ν/ν_o
0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
500	0.9887	0.9943	0.9421	0.9528	0.9761	0.9911	1.0402
1000	0.9774	0.9887	0.8869	0.9074	0.9526	0.9822	1.0824
1500	0.9661	0.9829	0.8344	0.8637	0.9293	0.9733	1.1269
2000	0.9549	0.9772	0.7845	0.8215	0.9064	0.9642	1.1737
2500	0.9436	0.9714	0.7369	0.7810	0.8837	0.9552	1.2230
3000	0.9323	0.9656	0.6918	0.7420	0.8614	0.9461	1.2750
3500	0.9210	0.9597	0.6489	0.7045	0.8394	0.9369	1.3298
4000	0.9097	0.9538	0.6082	0.6685	0.8176	0.9277	1.3877
4500	0.8984	0.9479	0.5696	0.6339	0.7962	0.9184	1.4488
5000	0.8872	0.9419	0.5329	0.6007	0.7751	0.9091	1.5133
5500	0.8759	0.9359	0.4983	0.5689	0.7542	0.8997	1.5816
6000	0.8646	0.9298	0.4654	0.5383	0.7337	0.8903	1.6538
6500	0.8533	0.9237	0.4344	0.5091	0.7135	0.8808	1.7303
7000	0.8420	0.9176	0.4050	0.4810	0.6936	0.8713	1.8113
7500	0.8307	0.9114	0.3773	0.4542	0.6739	0.8617	1.8972
8000	0.8194	0.9052	0.3511	0.4285	0.6546	0.8520	1.9884
8500	0.8082	0.8990	0.3264	0.4039	0.6356	0.8423	2.0852
9000	0.7969	0.8927	0.3032	0.3805	0.6168	0.8325	2.1881
9500	0.7856	0.8863	0.2813	0.3581	0.5984	0.8227	2.2976
10 000	0.7743	0.8799	0.2607	0.3367	0.5802	0.8128	2.4141
11 000	0.7517	0.8670	0.2232	0.2968	0.5448	0.7928	2.6707
12 000	0.7517	0.8670	0.1906	0.2535	0.5035	0.7928	3.1268
13 000	0.7517	0.8670	0.1628	0.2166	0.4654	0.7928	3.6608
14 000	0.7517	0.8670	0.1390	0.1850	0.4301	0.7928	4.2860
15 000	0.7517	0.8670	0.1188	0.1580	0.3975	0.7928	5.0181
20 000	0.7517	0.8670	0.0540	0.0718	0.2680	0.7928	11.0391

(Calculated using $T_0 = 288\ \mathrm{K}$, $\alpha = -0.0065\ \mathrm{K/m}$, $n = 5.256$,
and in Sutherland's formula, $A = 1.4665 \times 10^{-6}$, $B = 114\ \mathrm{K}$)

p – pressure, T - temperature, μ - dynamic viscosity

ρ - density, a - speed of sound, ν - kinematic viscosity, where $\nu = \mu / \rho$

Solution:

Standard sea level conditions,look at standard atmosphere table, hence ρ_0 = 1.225 kg/m^3

$$C_L = \frac{L}{0.5 \rho V^2 S}$$

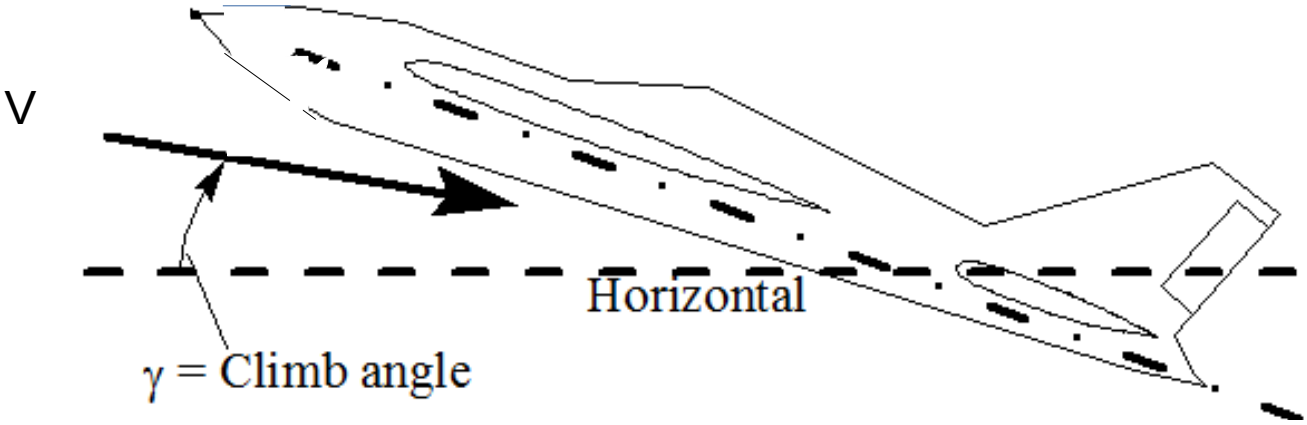
$$L = W$$

$$C_L = \frac{W}{0.5 \rho V^2 S}$$

$$1 = \frac{10000 * 9.81}{0.5 * 1.225 * V^2 * 10}$$

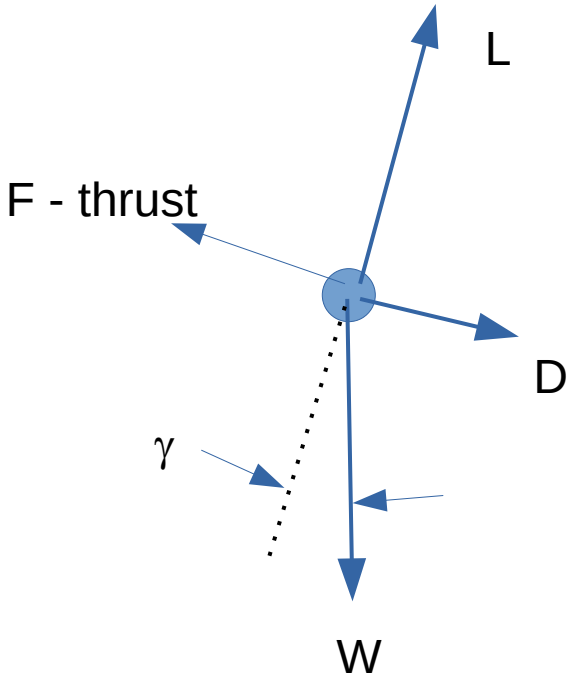
$$V = 126.49 (m/s)$$

Steady climb



L - Lift is always normal to V and not any axis of the aircraft
D - Drag is always parallel to V

Lets model the aircraft as a point:



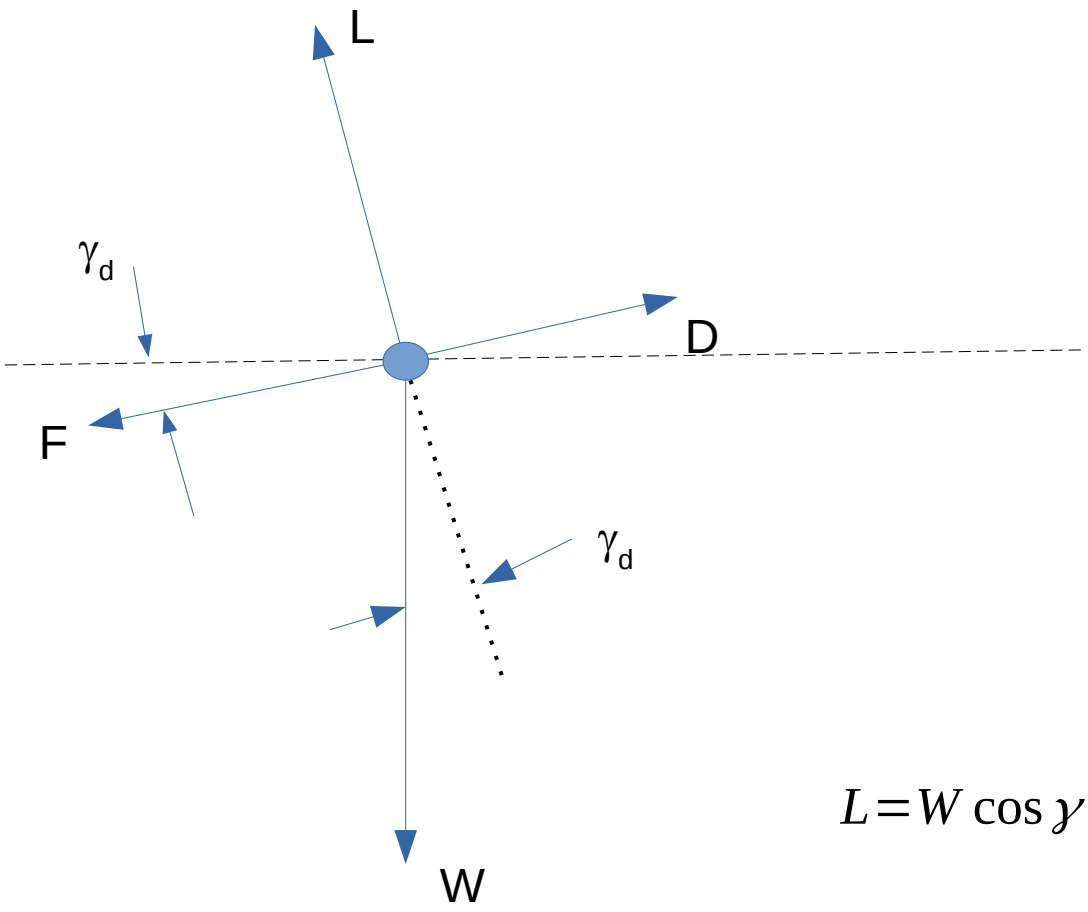
$$L = W\cos\gamma$$

$$F = D + W\sin\gamma$$

$$\sin\gamma = \frac{F-D}{W}$$

Thrust F is taken as in the opposite direction to Drag D
(no vectorized propulsion)

Steady descent



Horizontal

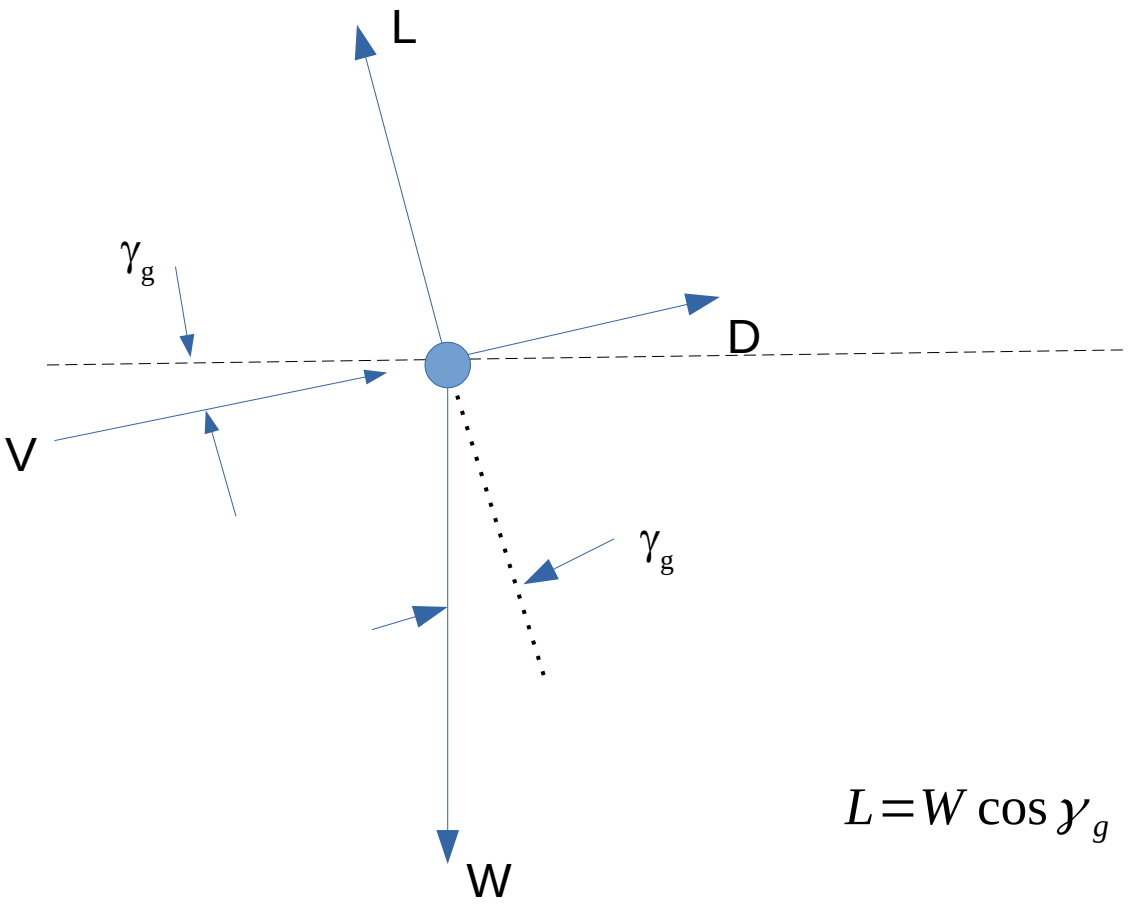
γ_d – descent angle.

$$L = W \cos \gamma_d$$

$$F + W \sin \gamma_d = D$$

$$\sin \gamma_d = \frac{D - F}{W}$$

Steady gliding, i.e. no thrust, F=0



Horizontal

γ_g – glide angle.

$$L = W\cos\gamma_g$$

$$W\sin\gamma_g = D$$

$$\sin\gamma_g = \frac{D}{W}$$

$$\tan\gamma_g = \frac{D}{L}$$

Hence gliders have high aerodynamic efficiency, i.e. L/D>>1, to get low γ_g, resulting in having wings of high AR.

In order to sustain a constant V, the glider loses potential energy (altitude) to convert it to kinetic energy to compensate for the loss of energy caused by the drag.

Example 2:

A glider has a 1:10 ratio of drag to lift. Calculate the glide angle.

Data:
D/*L* = 1/10

Question:
γ_g = ?

Solution:

$$\tan \gamma_g = \frac{D}{L}$$

$$\tan \gamma_g = \frac{1}{10}$$

$$\gamma_g = 5.71^0$$

Example 3:

The same glider was installed with an engine providing a thrust of 150% of the drag. Calculate the climb angle.

Data:
D/L = 1/10
F=1.5D

Question:
γ = ?

Solution:

$$\sin\gamma = \frac{F-D}{W}$$

$$\sin\gamma = \frac{1.5D-D}{W}$$

$$\sin\gamma = \frac{0.5D}{W}$$

But $W = L/\cos\gamma$ →

$$\tan\gamma = \frac{0.5D}{L}$$

$$\tan\gamma = 0.05$$

$$\gamma = 2.86^0$$

Second way: Assume γ << 1 (rad) and hence $W \simeq L$

$$\sin\gamma = \frac{0.5D}{L}$$

$$\sin\gamma = 0.05$$

$$\gamma = 2.86^0$$